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CoV2-Detect-Net: Design of COVID-19 prediction model based on hybrid DE-PSO with SVM using chest X-ray images,
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Abstract: For Covid-19 suspected cases, it is critical to diagnose them accurately and rapidly so that they can be isolated and provided with required medical care. A self-learning automation model will be helpful to diagnose the COVID-19 suspected individual using chest X-rays. AI based designs, which utilizes chest X-rays, have been recently proposed for the detection of COVID-19. However, these approaches are either using non-public database or having a complex design. In this study we have proposed a novel framework for real time detection of coronavirus patients without manual intervention. In our framework, we have introduced a 3-step process in which initially K-means clustering, and feature extraction is performed as a data pre-processing step. In the second step, the selected features are optimized by a novel feature optimization approach based on hybrid differential evolution algorithm and particle swarm optimization. The optimized features are then feed forwarded to SVM classifier. Empirical results show that our proposed model is able to achieve 99.34% accuracy. This shows that our model is robust and sustainable in diagnosis of COVID-19 infected individual.
Keywords: SARS-CoV-2; COVID-19; Differential evolution algorithm; Particle swarm optimization; Support Vector machine

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